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Chapter 2 Random variable and their probability distributions

Introduction

The sample space $\boldsymbol{S}$ is in general abstract. However, we may more-or-less naturally be able to define a correspondence **between the elements of $\boldsymbol{S}$** and **a set $\boldsymbol{R_S}$ of real numbers** (including perhaps a vector of real numbers).

- In this case, the probability measure defines a measure on that set of real numbers. This definition of the measure may be extended to all real numbers by defining the measure to be 0 for any real number $\boldsymbol{X} \notin \boldsymbol{R_S}$.

Introduction

- We will use the same symbol ***P*** **to denote the probability measure on** R_S as on the abstract sampling space ***S***. The real number ***X*** that takes on values according to this equivalent measure is called **a "random Variable".**
- We will loosely use the terms **probability distribution** or **sampling distribution** to refer to the probability measure and its possible functional descriptions.

Introduction

In this chapter we will tackle the concept of random variables, an important concept in probability and statistics.

- The concept of a random variable allows us to pass from the experimental outcomes themselves to a **numerical function of the outcomes.**

Introduction

- There are two fundamentally different types of random variables- **discrete random variables** and **continuous random variables**.
- In this chapter, we examine the basic properties and discuss the most **important examples** of discrete variables and continuous variables.

Example 1

Consider the experiment *E1* in 1.1:

Rolling a fair die, observing the points.

The sample space $S = \{1; 2; 3; 4; 5; 6\}$. We can give a function $X : S \rightarrow R^1$,

$$X(e) = e; \quad e = 1; 2; 3; 4; 5; 6$$

We can see that **the domain of *X* is the sample space *S***, and **the range of *X* is a subset of *R1*.** The function ***X*** represents the points occur when we rolling a fair die. We cannot say for sure the value that ***X*** will have on the next trial depends on the outcome of the die roll and that is something that we cannot predict exactly.

Example 2

For the experiment *E2* in 1.1:
Tossing a coin three times, observing conditions of frontage H and the reverse side T. The sample space

***S* = {HHH; HHT; HTH; THH; HTT; THT; TTH; TTT }**
and We can give a function X : S → R1,

$$X(e) = \begin{cases} 0; & \textit{if} \quad e = TTT \\ 1; & \textit{if} \quad e = HTT;\ THT;\ TTH \\ 2; & \textit{if} \quad e = HHT;\ HTH;\ THH \\ 3; & \textit{if} \quad e = HHH \end{cases}$$

It is easy to see that the function X represents the times of frontage H when we toss a coin three times.

Section 2.1

- Random variable
- Distribution function
- Examples

Random variable

Definition 2.1.1: Let E be a random trial, $\Omega = \{e\}$ is its sample space. If every $e \in \Omega$ has a corresponding real number $\boldsymbol{X(e)}$, then we have a function $\boldsymbol{X(e)}$ which is defined in sample space Ω, $\boldsymbol{X(e)}$ is called a random variable, marked as $\boldsymbol{r.v\ X}$.(random variable). Random variables usually are marked by capital letters $\boldsymbol{X}$, $\boldsymbol{Y}$, $\boldsymbol{Z}$ et al., or Greek letters ζ, η, ξ et al.

Note:

(1) Random variable is different from ordinary function

A random variable is a function, but it is fundamentally different from an ordinary function defined on the real number domain, whereas a random variable is defined on the sample space (the elements of the sample space are not necessarily real numbers)..

The observed value of X in the above examples varies at random. The randomness lies in the unpredictability of the specific value that X is going to take on next: **we don't know which of the values will occur on the next trial.** X is called a random variable or chance variable.

Random variable

(2) The value of random variable has certain probability distribution

The random variable has different values with the different results of the test. Since each result of the test has a certain probability, the value of the random variable ***also has a certain probability distribution.***

An ***r.v.*** is a function in definition, but it is rather different to the function in calculus. The only difference between an ***r.v.*** and a function in the usual calculus sense is that the domain of an ***r.v.*** is a sample space ***S***, which may be an abstract set , unlike the usual concept of a function, whose domain is a subset of $\boldsymbol{R^1}$ or of a Euclidean space of higher dimension.

Random variable

(3) There are **two common kinds of random variables**, **discrete and continuous**, and the difference between the two lies in the number of values in the sample space of the ***r.v.*.**. A discrete ***r.v.*** is one which can only assume a finite, or countably infinite, number of values. A continuous ***r.v.*** is one who can assume any value.

Random variable

Example 2.1.1: Tossing a dice, we can get two outcomes:

e1=(the frontage on the top)

e2=(the reverse side on the top)

Let X be the times that reverse on the top when tossing a dice, then we have：

$$\begin{array}{c} e_1 \\ e_2 \end{array} \xrightarrow{X(e)} \begin{array}{c} 0 \\ 1 \end{array} \quad \begin{array}{l} \rightarrow X(e_1)=0 \\ \rightarrow X(e_2)=1 \end{array}$$

i. e., X(e) is a random variable.

Random variable

Classification of random variables

- Random variables
  - Discrete
  - Non-discrete
    - Continuous
    - Others

(1) Discrete random variables: the possible values of a random variable are finite or infinite (can be listed).
Examples?
(2) **Continuous random variables**: The possible values of a random variable are among a continuous interval.
Examples?

Section 2.1

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